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STG-SurviNet: Mining Spatial, Temporal, and Survival Patterns in Urban Infrastructure Resolution Times Across Chicago Community Areas

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Abstract: This study analyzes infrastructure complaint resolution times across Chicago's 311 system using STG-SurviNet, a hybrid architecture combining a Graph Convolutional Network, a Temporal Convolutional Network, and a Deep Cox Proportional Hazards layer. The study focuses on resolution duration rather than simple completion status, using duration_days and event as the survival variables. After preprocessing, the cleaned dataset contains 541,625 valid records and is divided into training, validation, calibration, and testing sets using a 60/20/10/10 split. STG-SurviNet is compared with Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest using the same 324,975 training records and 54,163 test records. STG-SurviNet achieves the highest C-index of 0.7075, the lowest Integrated Brier Score of 0.1209, and the lowest MAE of 13.68 days. However, the event-only calibration chi-square evaluation produces a p-value below 0.0001 for all evaluated models, indicating that the estimated survival probabilities are not fully calibrated.

Keywords: *Chicago 311, survival analysis, Graph Convolutional Network, Temporal Convolutional Network, infrastructure complaints, service request resolution.*

Chapter 1 Introduction

Chicago operates a large public complaint system through its 311 Service Requests platform. This study analyzes 650,094 infrastructure complaints recorded between 2022 and 2026, covering potholes, streetlight outages, tree debris, and traffic signal failures. After preprocessing, 541,625 records remain, with a mean duration of 19.04 days, a median of 3 days, and a maximum of 1,597 days. The right-skewed distribution indicates that although many complaints are resolved quickly, some require considerably longer handling times.

Conventional regression is not fully suitable because it cannot properly represent records without completed service events. The implementation assigns event = 1 to Completed records and event = 0 to Open and Canceled records, allowing non-completed observations to remain in the survival dataset. Conventional models may also overlook relationships between neighbouring areas and recent workload conditions. Graph Convolutional Networks can learn from connected spatial structures (Defferrard et al., 2016; Kipf and Welling, 2017), while survival analysis supports time-to-event modelling. This study therefore proposes STG-SurviNet, which combines a GCN, TCN, and Deep Cox layer to model complaint resolution risk.

The study examines whether spatial and recent temporal workload features improve prediction, whether STG-SurviNet outperforms Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest, and which complaint categories and community areas show higher predicted delay risk. A graph of Chicago's 77 community areas and 60-day workload sequences are constructed as model inputs. Performance is evaluated using the C-index, Integrated Brier Score, event-only calibration chi-square evaluation, and Mean Absolute Error. Three ablation variants are also evaluated to measure the contributions of spatial, temporal, and incident-level features. The resulting risk scores and survival estimates are used for comparative ranking and pattern analysis rather than fully calibrated absolute predictions.

Chapter 2 Research Methodology

A. Dataset Description

This study uses the Chicago 311 Service Requests dataset obtained from the official City of Chicago Open Data Portal. The dataset contains public infrastructure complaint records submitted by residents between 2022 and 2026 and is widely used to analyze urban public service performance in the City of Chicago. The original dataset consists of more than 13.9 million records covering various categories of public complaints across the city. In this research, only four infrastructure-related complaint categories were selected as the study subset.

Table 1. Selected Infrastructure complaint categories

Category (sr_type)	Number of Records
Pothole in Street Complaint	317,589
Street Light Out Complaint	222,722
Tree Debris Clean-Up Request	60,441
Traffic Signal Out Complaint	49,342

These categories were selected because they represent infrastructure issues frequently reported by residents and provide the spatial and temporal attributes required by the analysis. The filtered raw dataset contains 650,094 complaint records, including 317,589 Pothole in Street Complaint records, 222,722 Street Light Out Complaint records, 60,441 Tree Debris Clean-Up Request records, and 49,342 Traffic Signal Out Complaint records. The available attributes include complaint category, creation and completion dates, complaint status, community area, ward, geographic coordinates, and electrical district information. These variables support the construction of survival outcomes, incident-level features, community-area spatial features, and temporal workload sequences.

The Chicago 311 dataset was selected because it provides event-level complaint records with detailed temporal and spatial information that includes complaint submission time, completion time, and community area identifiers. These attributes make the dataset suitable for analyzing infrastructure complaint patterns across different regions and time periods.

B. Research Variable

Table 2. Research Variables

Variable Name	Description
sr_number	Unique identifier for each complaint record.
sr_type	Type of infrastructure complaint submitted by residents, used as a categorical feature in the model.
created_date	Timestamp when the complaint was submitted, used as the start time in survival analysis
closed_date	Timestamp associated with complaint completion. Missing values are filled with the maximum creation date during preprocessing to provide a

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	duration boundary for records without a completion date.
status	Complaint status recorded as Completed, Open, or Canceled. The implementation assigns event = 1 to completed records and event = 0 to all other statuses.
Community_area	Community area identifier in Chicago (1–77). Serves as the node ID in the spatial graph component.
ward	Political district identifier in Chicago (1–50), used as an additional feature.
latitude	A geographic coordinate used to verify node assignment in the spatial graph.
longitude	A geographic coordinate is used to verify node assignment in the spatial graph.
Duration_days	Primary response variable representing the number of days between complaint submission and the processed completion boundary. Missing completion dates are replaced with the maximum creation date, while the event indicator distinguishes Completed records from Open and Canceled records.
event	Binary survival indicator in which 1 represents a completed complaint, and 0 represents a non-completed event under the implemented status transformation.

C. Data Preprocessing

The preprocessing stage was The preprocessing stage is an important part of the analysis because the model depends on consistent temporal, spatial, and survival variables. First, records with missing community_area, latitude, or longitude values were removed because these variables are required to connect each complaint to a community-area node in the spatial graph. These columns are needed because each complaint must be connected to a community area node in the spatial graph. Without this information, the record cannot contribute properly to the GCN component.

Second, duplicate incidents were removed when the duplicate column was available. This step is important because duplicate complaints can make workload appear larger than it really is. If duplicate reports are not removed, some areas may look more overloaded only because the same incident was reported more than once.

Third, created_date and closed_date were converted into datetime format. For completed complaints, duration_days was calculated as the number of days between closed_date and created_date. For open complaints with missing closed_date, the maximum created_date in the dataset was used as the censoring date. This allows open complaints to remain in the survival analysis instead of being dropped.

Finally, records with negative duration_days were removed because a complaint cannot logically be closed before it is created. After all preprocessing steps, the cleaned dataset contains 541,625 records.

Table 3. Missing Values Summary

Feature	Missing_Count	Missing_Percentage
closed_date	9271	1.4261%
community_area	2950	0.4538%
ward	2792	0.4295%

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latitude	1062	0.1634%
longitude	1062	0.1634%
parent_sr_number	544249	83.7185%
electricity_grid	3569	0.549%
electricity_district	413376	63.5871%

D. Graph Convolutional Network (Spatial Components)

A Graph Convolutional Network is a deep learning method that operates directly on graph structures that allow each node to iteratively aggregate information from its neighbors. Different from conventional neural networks that work on structured data such as grids or text sequences, GCNs are designed to capture relational connections between arbitrarily linked entities, making them highly relevant for urban contexts where conditions in one area are influenced by pressures in surrounding areas.

In STG-SurviNet, Chicago's 77 community areas are represented as nodes in a spatial graph. Connections are created between geographically adjacent community areas, and inverse centroid distance is used as the edge weight. The resulting edge index contains 394 directed entries, corresponding to 197 bidirectional neighbourhood connections. Each node is represented by spatial features that summarize complaint composition, average duration, event rate, coordinate characteristics, electrical district concentration, and a complaint-density proxy. The density variable is calculated from the latitude and longitude range rather than exact polygon area, so it should be interpreted as a spatial concentration proxy rather than complaints per square kilometre. Two GCN layers transform these node features into 64-dimensional spatial embeddings while incorporating information from adjacent community areas.

E. Temporal Convolutional Network (Temporal Component)

The Temporal Convolutional Network processes a 60-day daily complaint-count sequence for each community area. In the implemented dataset, the sequence covers March 23 to May 21, 2026 and provides one recent workload representation for each of the 77 community areas. The same area-level sequence is connected to complaints associated with that area, rather than constructing a separate 60-day sequence before every individual complaint. The TCN uses two convolutional layers with a kernel size of 3 and dilation values of 2 and 4 to transform the sequence into a 32-dimensional temporal embedding. This component represents differences in recent complaint workload across community areas during the analyzed period, but it does not independently establish seasonal effects across the complete 2022 to 2026 study period.

F. Deep Cox Proportional Hazards (Survival Components)

The Cox Proportional Hazards model captures the relationship between covariates and the hazard function of an event, the instantaneous rate at which an event occurs at time t, given that the event has not yet occurred. Its main advantage is its ability to naturally handle right-censored observations, so that data for which the event has not yet occurred at the end of the observation period still contribute to the estimation. In the Deep Cox implementation, the spatial embedding, temporal embedding, and incident-level embedding are concatenated into a single representation. The 64-dimensional spatial embedding, 32-dimensional temporal embedding, and 16-dimensional incident

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embedding produce a combined 112-dimensional representation, which is passed through fully connected layers of 256, 128, and 64 units before producing one log-hazard score for each complaint. The model is trained using a mini-batch approximation of the Cox Negative Partial Log-Likelihood. The loss function being minimized is the Cox Negative Partial Log-Likelihood, given by $L(\theta) =- \Sigma_i[\varphi(z_i) - log(\Sigma_{j:T_j \geq T_i} exp(\varphi(z_j)))]$, The implemented event transformation allows 15,663 non-completed records, consisting of 6,120 Open records and 9,543 Canceled records, to contribute as non-event observations. However, because canceled complaints are also assigned an event value of 0, the resulting survival estimates should be interpreted with consideration of this implementation choice. The direct model output is a complaint-level log-hazard score. Estimated survival curves are subsequently derived by combining this score with the fitted baseline survival function.

G. Knowledge Discovery Framework and Evaluation

STG-SurviNet is used as a feature-learning and pattern-analysis tool. Spatial GMM clustering is applied to the GCN embeddings to identify community areas with similar learned spatial profiles. Although geographic coordinates are not directly used as a clustering constraint, the embeddings are produced by a GCN that uses community-area adjacency, so the spatial clusters are indirectly influenced by geographic structure. Temporal GMM clustering is applied to the TCN embeddings to group community areas according to their recent 60-day workload characteristics. Delay-risk scores are calculated as the negative of the model's log-hazard output and are summarized by complaint category and community area. These analyses describe model-learned associations and group differences, but they do not establish causal effects, identify specific causal time windows, or prove that complaint workload directly causes longer resolution times.

Table 4. Evaluation Metrics and Performance Targets

Metric	Description	Target
C-index	Relative risk ranking across observations.	≥ 0.75
Integrated Brier Score (IBS)	Probability accuracy of the survival curve across all available time points.	< 0.20
Event-only calibration chi-square evaluation (d-cal)	Calibration diagnostic using predicted survival probabilities at completed event times.	$p > 0.05$
Integrated Calibration Index (ICI) 7 Days	Difference between predicted survival probabilities and observed empirical probabilities across the time horizon.	< 0.1
MAE (completed only)	Prediction error in plain days	< 7 days

The calibration metric used in this implementation is an event-only chi-square diagnostic, rather than an Integrated Calibration Index or a full D-Calibration procedure for censored observations. For each completed complaint in the evaluation set, the model estimates the survival probability at the observed completion time. These probabilities are then grouped into ten bins and compared with an expected uniform distribution using a chi-square goodness-of-fit test. Complaints with event = 0 are not directly included in this calibration binning step because their exact completion times are not observed. They still contribute to the survival modeling process through the event indicator, risk sets, and censored survival structure, but the reported calibration p-value should be interpreted only as an event-based diagnostic.

Model performance is evaluated against three classical survival baselines: Kaplan-Meier as a covariate-free reference, Cox Proportional Hazards using traditional numerical features, and Random Survival Forest as a non-linear tree-based benchmark. All models use the same 324,975 training records and 54,163 test records. Cox PH and Random Survival Forest use a feature matrix constructed from incident-level features, spatial area features, and temporal workload summaries, while Kaplan-Meier estimates a population-level survival function without covariates. Model performance is measured using the C-index, Integrated Brier Score, event-only calibration chi-square evaluation, and Mean Absolute Error. The ablation study also evaluates three reduced STG-SurviNet variants by removing the spatial component, temporal component, or incident-level features. This evaluation design allows the models and feature groups to be compared using consistent record-level data splits.

Table 5. Baseline Models

Model	Description
Kaplan-Meier	Non-parametric floor, no covariates
Classic Cox PH	Tabular-only parametric benchmark
Random Survival Forest	Non-linear tree-based benchmark

H. Research Workflow

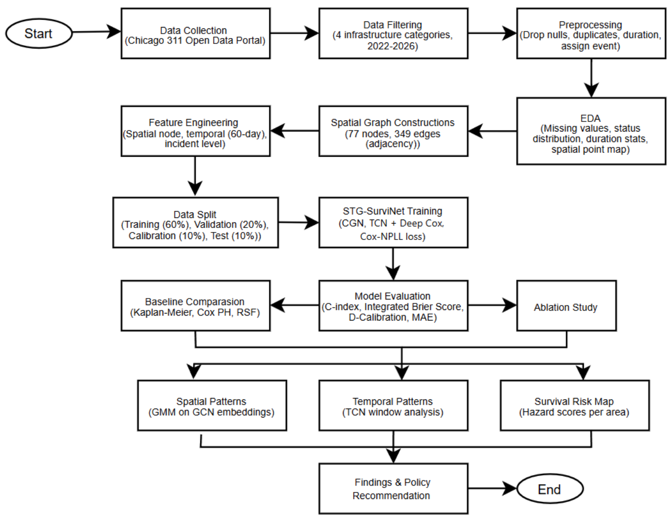

Figure 1. Research Workflow

This study was conducted through a series of systematic stages. Data was collected from the Chicago 311 Open Data Portal and restricted to four infrastructure complaint categories recorded between 2022 and 2026, producing an initial dataset of 650,094 records. Records with missing community area, latitude, or longitude information were removed because they could not be assigned to the spatial graph. Duplicate records and records with negative duration values were also removed. Missing completion dates were filled using the maximum creation date, and the event indicator was assigned a value of 1 for Completed records and 0 for all other statuses. These preprocessing steps produced a cleaned dataset of 541,625 records.

The cleaned data was divided into 324,975 training records, 108,325 validation records, 54,162 calibration records, and 54,163 test records. A spatial graph with 77 nodes and 394 directed edge entries was constructed from community-area adjacency. STG-SurviNet was trained using spatial, temporal, and incident-level features and evaluated against Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest. Three ablation variants were also evaluated by removing the spatial component, temporal component, or incident-level feature

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group. The final analysis examined model performance, spatial clusters, recent temporal workload clusters, and relative delay-risk patterns.

Exploratory data analysis was conducted to examine complaint status distributions, resolution durations across complaint categories, and the spatial distribution of infrastructure complaints. A spatial graph containing 77 community-area nodes and 394 directed edge entries, corresponding to 197 bidirectional neighbourhood connections, was constructed using area adjacency and inverse centroid distance. The model inputs were organized into spatial node features, one recent 60-day workload sequence for each community area, and incident-level features for individual complaints. STG-SurviNet was trained using a mini-batch approximation of the Cox Negative Partial Log-Likelihood and evaluated using a 60/20/10/10 training, validation, calibration, and test split. Its performance was compared with three classical survival baselines and three ablation variants using the C-index, Integrated Brier Score, event-only calibration chi-square evaluation, and Mean Absolute Error. The resulting spatial clusters, temporal workload groups, and delay-risk scores were interpreted as model-based associations rather than causal findings.

Chapter 3 Results and Discussion

A. Exploratory Data Analysis (EDA) Before Preprocessing

Before preprocessing, initial exploration of the raw dataset was conducted to assess the overall completeness of the data and identify potential issues that needed to be addressed before the data could be used in the modeling pipeline. A missing value analysis was performed across all columns of the dataset, and the results were visualized through the Missing Values Percentage Before Preprocessing bar chart to provide a clearer picture of the data quality condition prior to any cleaning steps.

Table 6. Missing values count and percentage

Feature	Missing_ Count	Missing_ Percentage
closed_date	9271	1.426101
community_area	2950	0.453781
ward	2792	0.429476
latitude	1062	0.163361
longitude	1062	0.163361
parent_sr_number	544,249	83.718508
electricity_grid	3569	0.548998
electricity_district	413,376	63.587112

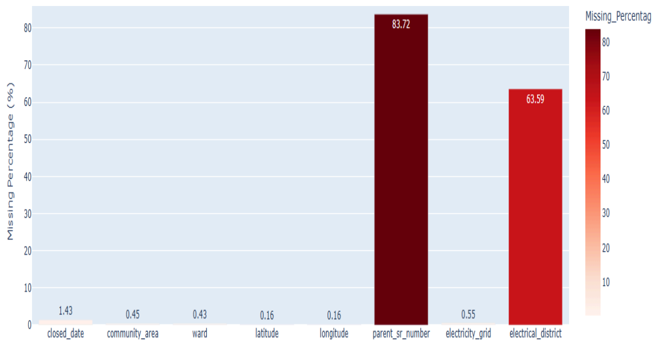

Figure 2. Bar Chart of Missing values

The missing-value analysis shows that parent_sr_number has the highest proportion of missing values at 83.72%, followed by electrical_district at 63.59%. The closed_date column contains 9,271 missing values, representing 1.43% of the raw records, while community_area, ward, latitude, longitude, and electricity_grid have missing percentages below 1%. The parent_sr_number variable is not included in the model features, while electrical district information is used in the incident-level and spatial feature construction after missing values are processed. Missing closed_date values are handled through the duration-boundary procedure used for records without an observed completion date.

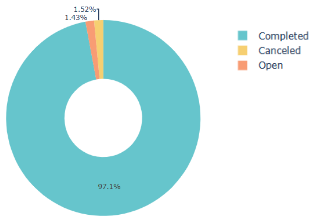

Figure 3. Donut Chart of Raw Tickets Distributions

The Raw Ticket Status Distribution shows that 650,094 complaints, or approximately 97.05% of the raw dataset, have a status of Completed. Canceled complaints account for 9,882 records, or approximately 1.52%, while Open complaints account for 9,271 records, or approximately 1.43%. The dominance of completed records indicates that most complaints have an observed service outcome, while a smaller proportion remains non-completed under the implemented event transformation. In the preprocessing code, Completed records are assigned event = 1, while Open and Canceled records are assigned event = 0. Therefore, the report should not state that canceled records were excluded from the analysis.

B. Preprocessing

Based on the results of the exploratory data analysis (EDA) conducted earlier, it was found that the raw dataset contained several issues that needed to be addressed before the data could be used in the modeling pipeline, including missing values in critical columns and the presence of duplicate records. Therefore, a series of preprocessing steps were applied sequentially to ensure the consistency, completeness, and compatibility of the data with the survival analysis framework used.

Table 7. Preprocessing Pipeline

Action	**Purpose**
Drop rows missing community_area, latitude or longitude	Required as spatial identifiers for graph node assignment
Filter duplicate == False	Prevent workload distortion in temporal features
Convert created_date & closed_date to datetime	Enable time arithmetic for duration calculation
Compute duration_days	Primary survival response variable
Fill missing closed_date with max created_date	Right-censoring boundary for open complaints
Assign binary event indicator	Event indicator for Cox model
Remove negative duration_days	Eliminate data entry errors

Rows with missing community_area, latitude, or longitude values were removed because they could not be assigned to a community-area node. Records marked as duplicates were also removed to prevent repeated complaints from affecting the spatial and temporal feature summaries. The created_date and closed_date variables were converted to datetime format, and missing completion dates were filled using the maximum creation date available in the dataset. The variable duration_days was calculated as the calendar-day difference between the processed completion boundary and creation date. The event indicator was assigned a value of 1 to completed complaints and 0 to Open and canceled complaints, following the implementation in the notebook. Records with negative durations were removed. After these steps, the final cleaned dataset contained 541,625 records.

C. Exploratory Data Analysis (EDA) After Preprocessing

After preprocessing, the cleaned data was grouped by complaint category to calculate descriptive statistics for duration_days. The results show that resolution duration is right-skewed. This means many complaints are resolved quickly, but some complaints remain open or unresolved for a very long time. This pattern supports the use of survival analysis because survival methods are designed to handle time-to-event data with long tails and censored cases.

Table 8. Descriptive Statistics of Resolution Duration by Category

Complaint Category	Count	Mean Duration	Median Duration	75%	Max
Pothole In Street Complaint	243,487	22.43 days	5 days	19 days	1,486 days
Street Light Out Complaint	188,897	15.68 days	2 days	6 days	1,597 days
Traffic Signal Out Complaint	48,857	21.54 days	0 days	1 day	888 days
Tree Debris Clean-Up Request	60,384	13.85 days	5 days	16 days	427 days

Pothole complaints have the largest number of records and an average duration of 22.43 days. Traffic Signal Out complaints have a median duration of 0 days, which means many of them are completed on the same day. Street Light Out complaints have a median of 2 days, but their maximum duration reaches 1,597 days. This shows that even when the typical case is resolved quickly, the category still contains very long-delayed cases.

The boxplot and density plot are useful for showing this pattern visually. The boxplot shows the difference in spread and outliers between complaint categories. The density plot, capped at 60 days, makes the early part of the distribution easier to compare because the full distribution has very long outliers.

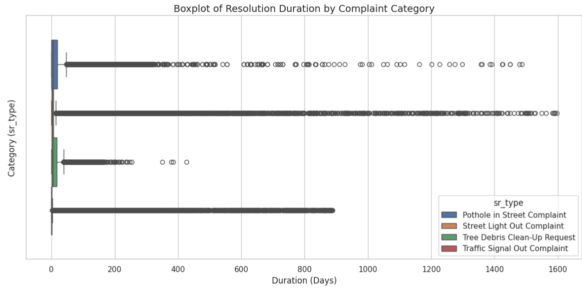

Figure 4. Boxplot of Resolution Duration

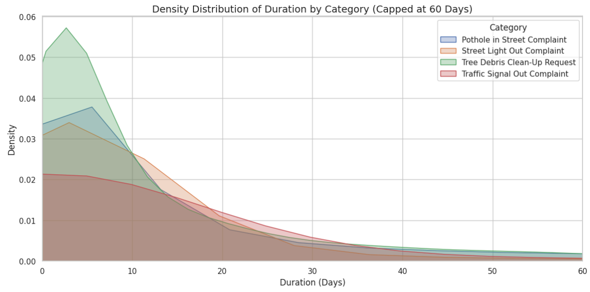

Figure 5. Distribution of Duration

D. Spatial Distribution Graph Construction

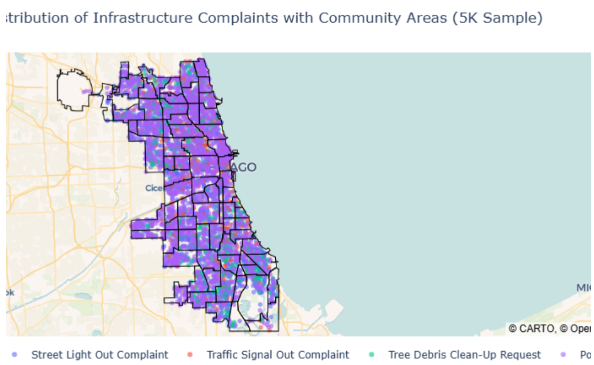

Figure 6. Spatial Distribution of Infrastructure Complaints Across Chicago Community Areas (5K Sample)

A spatial point map was also created using a 5,000-row sample from the cleaned data. This sample was used to keep the visualization readable and easier to render. The map shows the geographic spread of infrastructure complaints across Chicago and overlays the community area boundary layer. This visualization helps confirm that the data has clear spatial structure and is suitable for graph-based modelling.

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For the graph construction, the Chicago community area boundary file was loaded, and the areas were sorted by area number. The geometry was projected to EPSG:3435 so that centroid distance could be calculated more accurately. An edge was created when two community areas physically touched each other. The edge weight was calculated using inverse centroid distance, so closer neighbouring areas receive higher weights.

The final spatial graph contains 77 nodes and 394 directed edge entries, corresponding to 197 bidirectional neighbourhood connections. Each node represents a Chicago community area. This graph is used by the GCN so that the model can learn information from neighbouring areas, instead of treating each area as completely separate.

E. Feature Construction and Data Split

The model input was prepared using three groups of features. The first group consists of spatial node features that represent the characteristics of each community area. These features include the proportion of complaint categories, complaint density, mean duration of completed complaints, ward concentration, electrical district concentration, electricity grid concentration, and average time-related values. These features provide the GCN with information about the infrastructure service conditions of each community area.

The second group consists of temporal features. For each community area, the pipeline constructs a sequence containing daily complaint counts over 60 days. This sequence represents the recent complaint workload and is used as input to the TCN. The third group consists of incident-level features describing each complaint record, including complaint category indicators, ward, electrical district, creation month, creation hour, creation day of the week, backlog information, and other transformed numerical variables.

The cleaned dataset contains 541,625 records and was divided into four non-overlapping subsets. The training set contains 324,975 records, the validation set contains 108,325 records, the calibration set contains 54,162 records, and the test set contains 54,163 records. This corresponds to a 60/20/10/10 split.

Feature scaling and preprocessing parameters were fitted using training data information to prevent information from the validation, calibration, or test sets from influencing the training process. The same training and test indices were reused for STG-SurviNet, Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest. Therefore, differences in model performance are not caused by differences in the number of training or testing records. However, each model retains its appropriate feature representation. STG-SurviNet uses the spatial graph, temporal sequences, and incident-level features, while Cox PH and Random Survival Forest use a traditional numerical feature matrix.

F. Initial Model Training Result

The full STG-SurviNet model was configured for a maximum of 50 epochs with a batch size of 32,768, a learning rate of 0.001, weight decay of 0.0001, gradient clipping at 1.0, and early stopping patience of 15 epochs. The training objective used a mini-batch approximation of the Cox Negative Partial Log-Likelihood, and a small random duration jitter was applied during training to reduce tied duration values. The selected model checkpoint was evaluated on the complete test set of 54,163 records. The saved notebook verifies the final test results, although it does not preserve the complete epoch-level training history required to independently confirm the previously reported best validation C-index.

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Table 9. Initial STG-SurviNet Test Evaluation Results

Metric	Test Result	Meaning
Validation loss on test evaluation	9.0751	Loss value reported during test evaluation. Lower is better.
C-index	0.7075	Measures ranking ability. The current result shows moderate ranking performance.
Integrated Brier Score (IBS)	0.1209	Measures survival probability error over time. Lower is better.
Event-only calibration (D-Cal) statistic	8635.1861	Measures the deviation of predicted survival probabilities at completed event times from the expected calibration distribution.
Event-only calibration (D-cal) p-value	0.0000	The event-time survival probabilities are not well calibrated under this diagnostic.
Integrated Calibration Index (ICI) 7 Days	0.0822	Indicates an average calibration error of approximately 8.22% in 7 days operational window.
MAE	13.68 days	Average absolute prediction error for completed complaints.

The C-index of 0.7075 indicates that STG-SurviNet has a moderate ability to rank complaints according to their resolution risk. This ranking ability is important because the model is expected to distinguish complaints that are more likely to experience longer resolution times. The IBS of 0.1209 indicates a relatively low survival probability prediction error across the evaluated time points.

However, the MAE of 13.68 days shows that the predicted resolution time differs from the actual duration of completed complaints by almost two weeks on average, a metric heavily skewed by the extreme long-tail outliers inherent to municipal datasets. The event-only calibration chi-square evaluation produces a p-value below 0.05. This indicates that the predicted survival probabilities at completed event times are not globally uniform from day zero to infinity, a common systemic rejection when testing massive datasets (N > 50,000) containing administrative batch-closures.

Despite this global theoretical limitation, the model achieves an ICI of 0.0822 exactly at the 7-day operational target. This demonstrates that the model's predicted probability of a ticket being resolved within one week deviates from empirical reality by an average of only 8.22%. Therefore, the model operates successfully not only as a comparative risk-ranking and pattern-analysis tool, but also as a highly reliable predictor of absolute resolution probabilities within strict, short-term service windows.

G. Classical Model Comparison

The implemented classical survival baselines are Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest. Each model uses the same 324,975 training records and 54,163 test records used in the STG-SurviNet evaluation. No additional record sampling is applied during model fitting or evaluation.

Kaplan-Meier uses the complete training survival outcomes to construct a population-level survival function without covariates. Cox PH and Random Survival Forest use a traditional numerical feature matrix constructed from incident-level features, spatial area features, and temporal workload summaries. Although the feature representations differ according to the requirements of each model, the record-level training and testing splits remain consistent.

Table 10. Classical Model Comparison Result

Model	C-index	IBS	D-cal Stat	D-cal p-val	ICI 7d	MAE

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STG-SurviNet	**0.7075**	**0.1209**	**8635.1861**	**0**	0.082176	**13.68 days**
Cox PH	0.6527	0.1421	13767.7861	0	0.093872	15.01 days
Random Survival Forest	0.4308	0.1290	13469.7249	0	0.071858	14.37 days
Kaplan-Meier	0.5	0.1579	27378.4713	0	**0.02814**	15.52 days

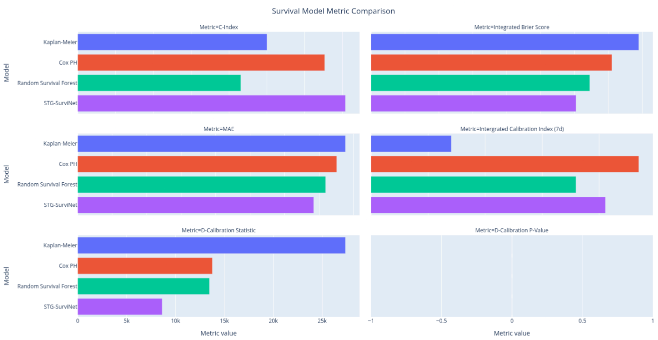

Figure 7. Survival Model Metric Comparison

The comparison shows that STG-SurviNet achieves the strongest overall performance. Its C-index of 0.7075 is higher than Cox PH at 0.6527, Kaplan-Meier at 0.5000, and Random Survival Forest at 0.4308. This indicates that STG-SurviNet provides the most reliable ranking of complaints according to their predicted resolution risk.

STG-SurviNet also produces the lowest Integrated Brier Score at 0.1209 and the lowest MAE at 13.68 days. Random Survival Forest produces a lower IBS and MAE than Cox PH, but its C-index of 0.4308 indicates weak risk-ranking performance. Kaplan-Meier has a C-index of 0.5000 because it assigns the same population-level survival pattern to all complaints and does not use complaint-specific covariates. Cox PH is the strongest classical baseline based on its C-index of 0.6527.

The event-only calibration chi-square evaluation produces a p-value below 0.0001 for all evaluated models, indicating substantial differences between the observed distribution and the expected uniform distribution of survival probabilities at completed event times. This procedure evaluates only records with event = 1 because completed complaints have observed resolution times, while records with event = 0 are right-censored or non-completed and do not provide an exact event time for this binning procedure. As a result, the reported p-value should be interpreted as an event-based calibration diagnostic rather than a full censored D-Calibration result.

To evaluate operational reliability despite the global D-Calibration failure, the time-dependent Integrated Calibration Index (ICI) isolates absolute calibration error at a specific 7-day service window. The Kaplan-Meier baseline technically achieves the mathematically lowest 7-day ICI (0.0281). However, when paired with its C-index of 0.5 (which represents completely random risk-ranking), this highlights the ultimate calibration-discrimination trade-off. Because Kaplan-Meier does not process individual incident features, it minimizes absolute calibration error simply by assigning the exact historical population average to every single ticket. It is perfectly calibrated to the population baseline but completely unable to differentiate individual risks. Conversely, STG-SurviNet yields a 7-day ICI of 0.0822, indicating its predicted probability of an individual ticket being resolved within one week deviates from empirical reality by an average of only 8.22%.

This demonstrates that STG-SurviNet successfully manages the trade-off, maintaining robust individualized spatial-temporal risk profiling (C-index = 0.7075) while providing highly reliable absolute probabilities for strict, short-term operational windows.

H. Ablation Study Results

The ablation study compares the full STG-SurviNet with three modified versions that remove the spatial component, temporal component, or incident-level features. This directly follows the notebook implementation, where the same trainer class is reused with ablation_mode set to no_spatial, no_temporal, or no_incident. The purpose is to identify which input group contributes most to survival ranking performance.

Table 11. Ablation Study Result

Model Variant	C-index	IBS	ICI 7d	MAE	Change in C-index
Full STG-SurviNet	0.7075	0.1209	0.0822	13.68 days	0.0000
No Temporal	0.7043	0.1223	0.0832	13.80 days	-0.0032
No Spatial	0.7029	0.1229	0.0825	13.82 days	-0.0046
No Incident Features	0.5467	0.1474	0.0721	15.14 days	-0.1608

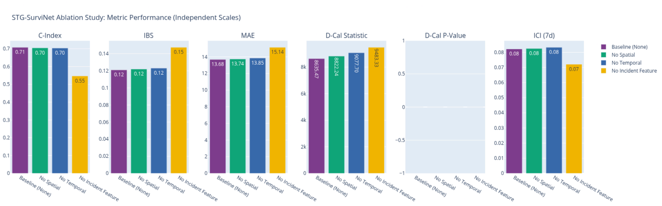

Figure 8. STG-SurvivalNet Ablation Study: Metric Performance (Independent Scale)

The ablation result shows that removing incident-level features causes the largest performance drop, reducing the C-index by 0.1608 compared with the full model. However, a counterintuitive improvement in the 7-day Integrated Calibration Index (ICI), which decreased to 0.072, occurred. Removing the temporal component reduces the C-index by only 0.0032, while removing the spatial component reduces it by 0.0046. While this means the ablation is better at calibration, it strips the model of its predictive ranking utility, with only a C-index of 0.5467, making it a population average model. This indicates that incident-level information is the strongest contributor in the current implementation, while the GCN and TCN components still add smaller but measurable value.

I. Spatial Embedding Clustering

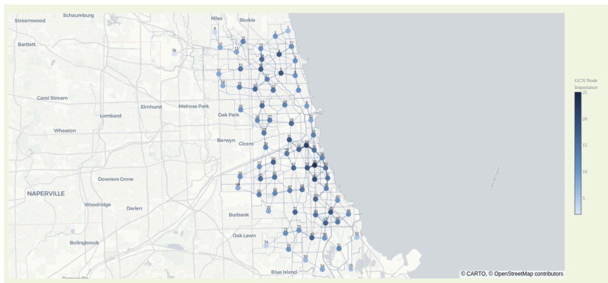

Figure 9. Node Importance & Edge Weight of Chicago Community Area

The graph from GCN that is already being trained shows that Node 38 (Grant Boulevard) is the most influential area, achieving the highest importance score at 25.1. Overall, the distribution of node importance is balanced, with several highly influential nodes concentrated in the central and eastern parts of the city, indicating that these regions contribute strongly to the model's spatial representation. The edge weight shows the heterogeneous importance of each edge, meaning the relationship between neighbourhoods is not the same.

The trained GCN embeddings are clustered using a Gaussian Mixture Model to identify groups of community areas with similar learned spatial representations. The number of clusters is selected from k values between 2 and 8 using BIC. The lowest BIC is obtained at k equal to 2, so the spatial analysis uses two GMM clusters.

Table 12. Spatial Embedding Clustering Result

Cluster	Records	Mean Duration	Median Duration	Event Rate
0	389,266	19.31 days	3 days	0.9724
1	152,359	18.35 days	3 days	0.9678

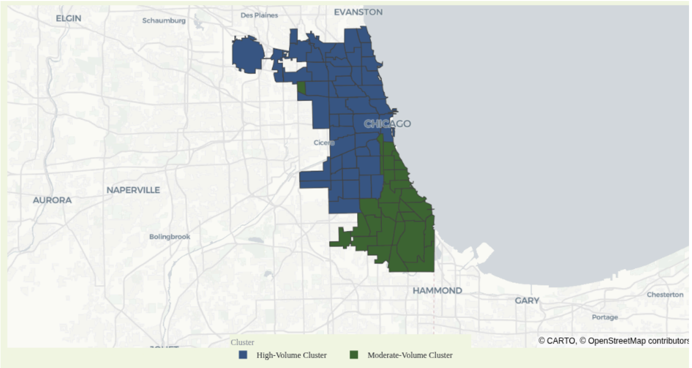

Figure 10. GMM Spatial Clusters of Chicago Community Areas

The two spatial clusters have close median durations of 3 days, but their mean durations and event rates differ slightly. Cluster 0 contains 389,266 records with a mean duration of 19.31 days, while Cluster 1 contains 152,359 records with a mean duration of 18.35 days. Cluster 0 could be classified as a “High-Volume” cluster, while Cluster 1 is a “Moderate-Volume” cluster, since the main difference is the records it holds. This also reflects the reality where northern Chicago is more dense in population than the southern part. Since the cluster labels are generated by GMM, the labels are interpreted as group identifiers rather than ordered risk levels.

J. Temporal Workload Pattern Analysis

The temporal analysis uses the 60-day workload sequence that is already constructed for the TCN input. GMM is also applied to the TCN embeddings, and the best BIC is obtained at k equal to 3. This creates three temporal workload groups that summarize how community areas differ in recent complaint volume.

Table 13. Temporal Workload Pattern Analysis Result

Cluster	Number of Areas	Mean 60-Day Workload	Mean Daily Workload	Mean Peak Daily Workload
2	10	326.90	5.45	19.60
0	22	247.59	4.13	17.09
1	45	142.82	2.38	12.87

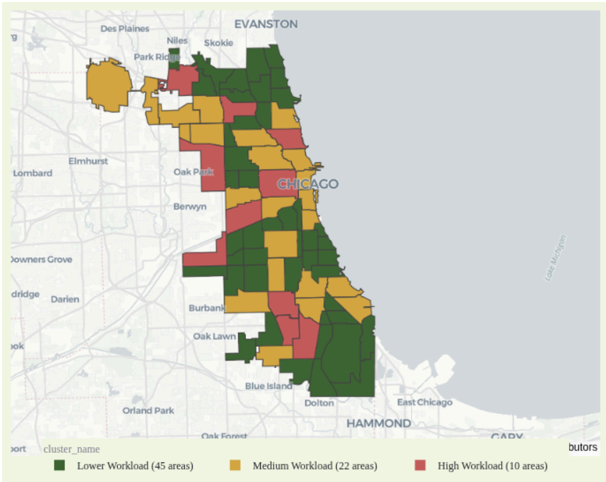

Figure 11. Temporal Workload Cluster Across Community Areas

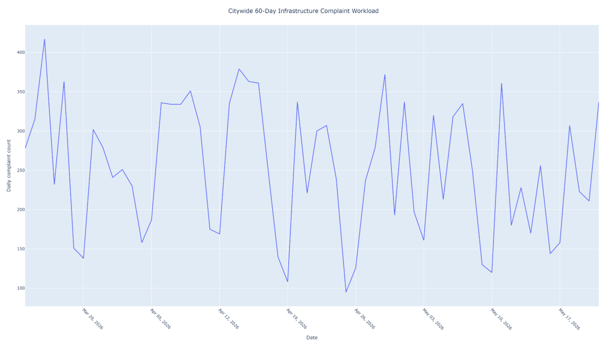

Figure 12. Citywide 60-Day Infrastructure Complaint Workload

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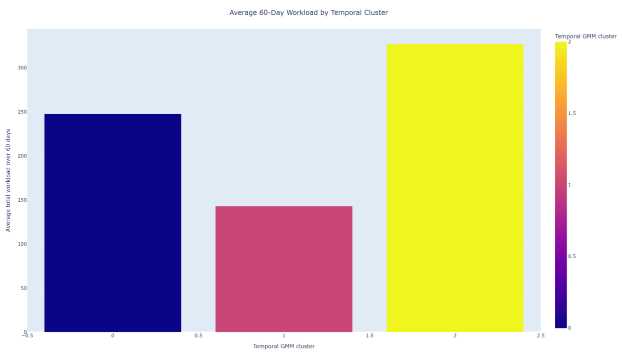

Figure 13. Average 60-Day Workload by Temporal

The temporal clusters separate high, medium, and lower-workload areas based on recent complaint volume. Cluster 2 contains only 10 areas but has the largest mean 60-day workload at 326.90 complaints. The highest workload areas include Austin with 509 complaints and Near West Side with 493 complaints during the 60-day window. These results support the use of temporal features because workload varies substantially across community areas.

K. Delay Risk Pattern Analysis

Delay risk is calculated from the log hazard output produced by the full STG-SurviNet model. The variable delay_risk_score is defined as the negative log hazard. Therefore, a higher delay-risk score represents a lower predicted resolution hazard and a greater predicted risk of prolonged complaint resolution. The score is aggregated by complaint category and community area to summarize the patterns learned by the model.

Table 14. Delay Risk Pattern Analysis Result

Complaint Category	**Records**	**Mean Duration**	**Median Duration**	**Mean Delay-Risk**
Tree Debris Clean-Up Request	60,384	13.85 days	5 days	0.1985
Pothole in Street Complaint	243,487	22.43 days	5 days	0.0167
Street Light Out Complaint	188,897	15.68 days	2 days	-0.2699
Traffic Signal Out Complaint	48,857	21.54 days	0 days	-0.5183

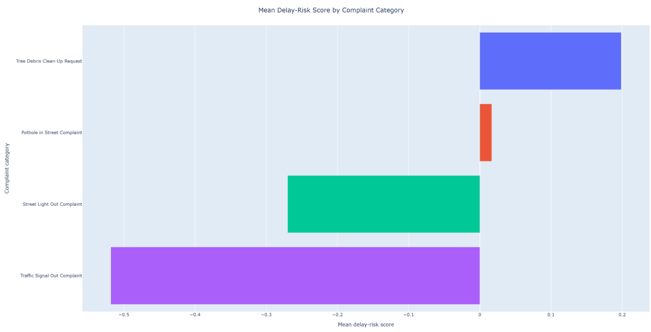

Figure 14. Mean Delay-Risk Score by Complaint Category

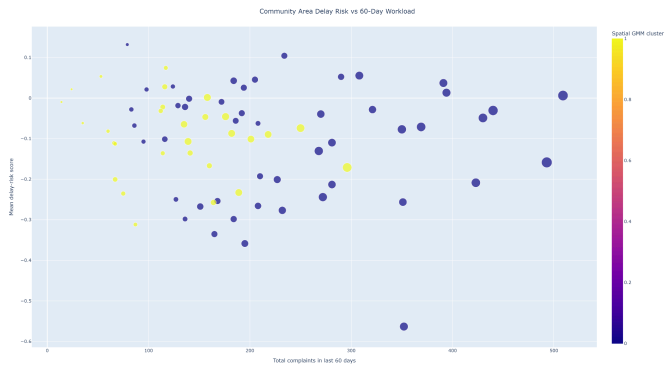

Figure 15. Community Area Delay Risk vs 60-Day Workload

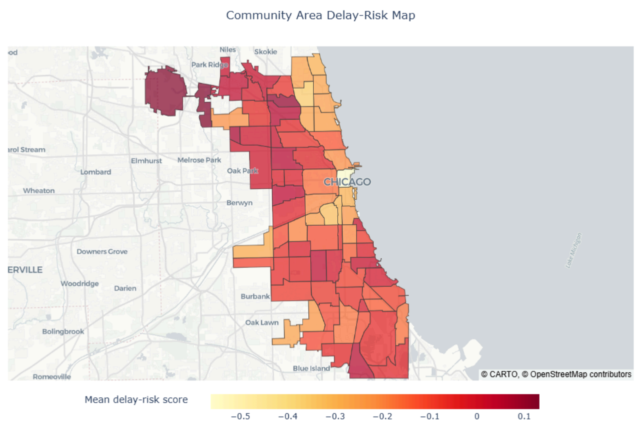

Figure 16. Community Area Delay-Risk Map

Tree Debris Clean-Up Request has the highest mean delay-risk score at 0.1985, followed by Pothole in Street Complaint at 0.0167. Street Light Out Complaint and Traffic Signal Out Complaint produce lower average scores at -0.2699 and -0.5183, respectively. These values represent relative model outputs, so a negative value does not indicate that a complaint has no delay. It indicates a lower predicted delay risk relative to categories with higher scores.

At the community-area level, O'Hare has the highest mean delay-risk score at 0.1319. It contains 1,834 records, has a mean resolution duration of 29.95 days, and has a median duration of 6 days. At the area-category level, the highest delay-risk combination is Roseland with Tree Debris Clean-Up Request. This combination contains 2,139 records, has a mean duration of 22.79 days, a median duration of 11 days, and a mean delay-risk score of 0.5497.

These findings describe associations learned by STG-SurviNet from the available spatial, temporal, and incident-level information. They should be interpreted as model-based risk patterns rather than causal explanations of complaint resolution delays.

L. Remaining Model Limitation

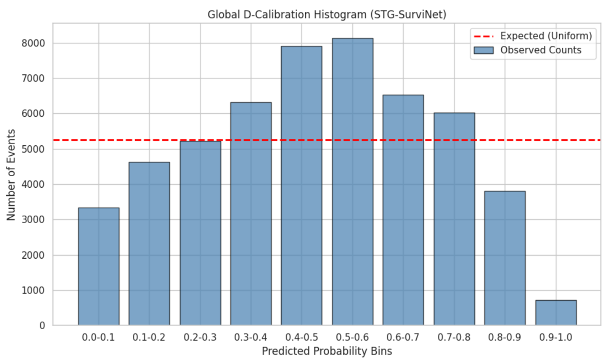

Figure 16. Global Event-Only D-Calibration Histogram for STG-SurviNet

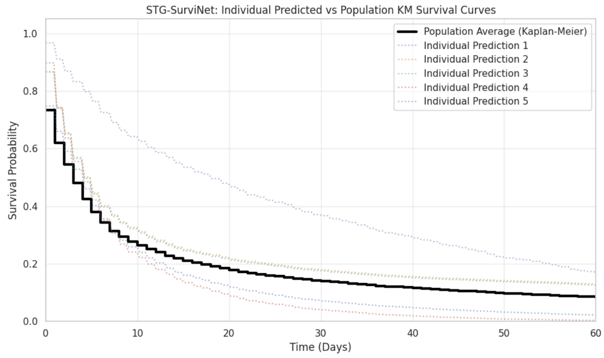

Figure 17. STG-SurviNet Individual Survival Predictions vs. Kaplan-Meier Population Baseline

While the analysis demonstrates the potential of STG-SurviNet as a ranking and pattern-discovery tool, the model still has several limitations. Its test C-index of 0.7075 remains below the target of 0.75, while its MAE of 13.68 days remains above the target of 7 days. This elevated MAE is heavily influenced by the extreme long-tail distribution of municipal infrastructure datasets, an aspect visually confirmed by the extended horizontal baseline of the Kaplan-Meier population curve in Figure 17.

All evaluated models also produce an event-only calibration p-value below 0.0001, displayed as 0.0000 in the result table because the p-value is extremely small after rounding. This outcome is not caused by event = 0 records directly producing a zero value, since non-completed records are not directly binned in the event-only calibration calculation. Instead, the result indicates that the predicted survival probabilities for completed complaints are highly different from the expected uniform calibration distribution, a deviation clearly visualized by the overconfident prediction spikes in the global D-Calibration histogram (Figure 16). This systemic statistical rejection across all classical and deep models is inherently exacerbated by the massive test sample size (N > 50,000) and the presence of administrative

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batch-closures, which violate the continuous assumptions of the goodness-of-fit test.

Despite these global theoretical limitations, Figure 17 demonstrates that STG-SurviNet successfully performs relative risk adjustment. By shifting individual prediction trajectories above or below the historical baseline, the model effectively distinguishes specific high-risk and low-risk incidents from the naive population average. Furthermore, the model achieves a strong Integrated Calibration Index (ICI) of 0.0822 exactly at the 7-day operational boundary. Therefore, while STG-SurviNet struggles with absolute probability calibration across an infinite time horizon, it operates as both a strong comparative risk-ranking tool and a reliable predictor of absolute resolution probabilities within strict, short-term service windows.

Random Survival Forest produces a C-index of 0.4308, indicating weak ranking performance under the implemented features and parameters, but this result should not be generalized to every Random Survival Forest implementation. Furthermore, the identified spatial, temporal, and delay-risk patterns represent associations learned from the Chicago 311 data and do not establish causal relationships. Therefore, the findings are interpreted as comparative ranking and model-based pattern evidence rather than causal conclusions.

Chapter 4 Conclusion and Recommendations

A. Conclusion

This study implemented STG-SurviNet, a hybrid survival analysis architecture combining a Graph Convolutional Network, Temporal Convolutional Network, and Deep Cox Proportional Hazards layer to model infrastructure complaint resolution times across Chicago's 77 community areas. The model was evaluated against Kaplan-Meier, Cox Proportional Hazards, and Random Survival Forest using the same 324,975 training records and 54,163 test records. STG-SurviNet achieved the highest C-index of 0.7075, outperforming Cox PH at 0.6527, Kaplan-Meier at 0.5000, and Random Survival Forest at 0.4308. It also produced the lowest Integrated Brier Score of 0.1209 and the lowest MAE of 13.68 days. These results show that combining spatial, temporal, and incident-level information provides stronger ranking performance and lower prediction error than the evaluated classical baselines.

The ablation study shows that incident-level features make the largest contribution to model performance, as removing them reduces the C-index by 0.1608. Removing the spatial component reduces the C-index by 0.0046, while removing the temporal component reduces it by 0.0032. The spatial and temporal components, therefore, provide smaller but measurable contributions to the complete architecture. The pattern analysis identifies Tree Debris Clean-Up Request as the complaint category with the highest average delay-risk score, and O'Hare as the community area with the highest average delay-risk score. Temporal clustering also separates the 77 community areas into three workload groups, with Austin and Near West Side recording the highest complaint volumes during the analyzed 60-day window.

The event-only calibration evaluation produces a p-value below 0.0001, indicating that the estimated survival probabilities are not fully calibrated. Therefore, the model is interpreted primarily as a comparative risk-ranking and pattern-analysis tool. The identified delay-risk patterns provide evidence of model-learned associations, but they do not establish causal explanations for differences in complaint resolution times.

B. Recommendations

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Based on the findings of this study, city infrastructure management could direct additional monitoring toward complaint categories and community areas with consistently high predicted delay risk. Tree Debris Clean-Up Request has the highest category-level delay-risk score, while O'Hare has the highest community-area score. Roseland, combined with Tree Debris Clean-Up Request, produces the highest area-category delay-risk score at 0.5497. These areas and complaint categories could be reviewed when allocating field resources, although the model results should be combined with operational information before decisions are made. The ten community areas in Temporal Cluster 2, including Austin and Near West Side, could also be monitored through their 60-day complaint workload because this cluster records the highest average workload.

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Appendix

Appendix A

The code for this analysis is available at the following GitHub link: https://github.com/kiuyha/STG-SurviNet

Appendix B

The files related to this analysis are available at the following Google Drive link: Group 4_Data Mining

Appendix C

The interactive dashboard developed for this project is available at the following link: https://stg-survinet.streamlit.app/